INSTITUTE OF DISTANCE LEARNING

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ME 392

INDUSTRIAL ENGINEERING AND ERGONOMICS

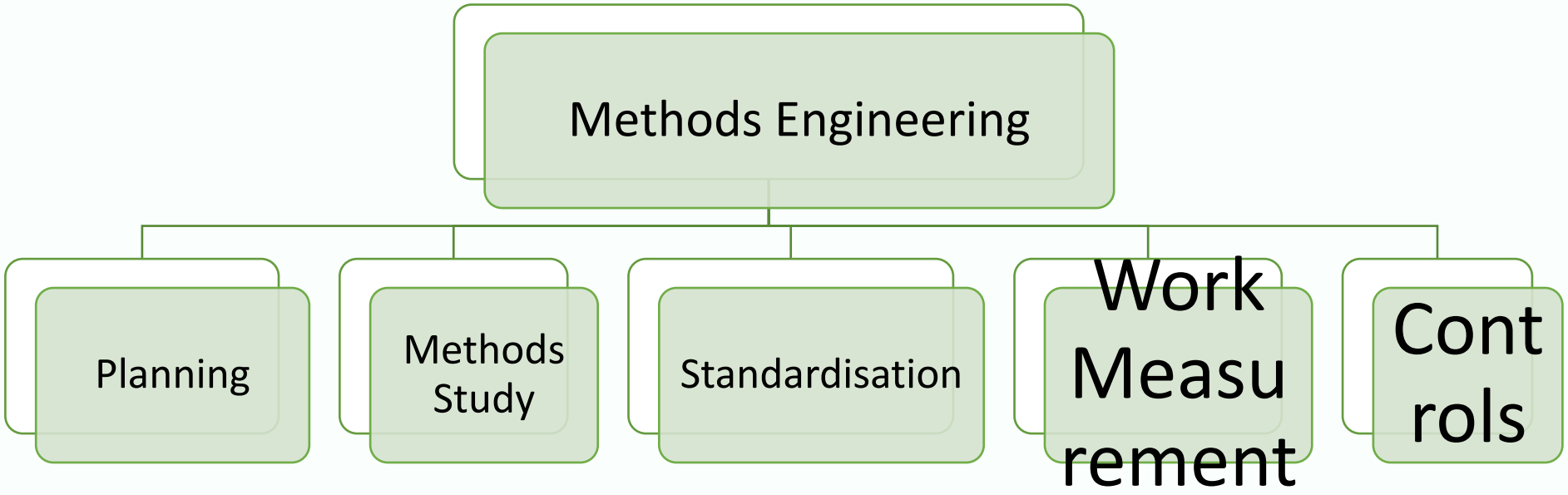
UNIT 2: METHODS ENGINEERING

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- Methods engineering includes five activities:



Planning

Identifying and understanding the problem & setting the objectives

- The problem may be:
  - low productivity,
  - high cost,
  - inefficient methods
- The objective is the desired improvement or new method design.
- Possible objectives are to increased productivity, reduced labor content and cost, improved safety, or developed new method or new operation.

Planning -2

Invariably the most profitable jobs to study are those with:

- the most repetition,
- the highest labour content (human work as distinguished from mechanical or process work),
- the highest labour cost,
- or the longest life span.

Methods Study

Analyze the Problem

- Collect data on the problem being studied & analyse activities
- Activities often used in this step include the following:
  - Identify the basic functions of the operation.
  - Gather background information.
  - Observe the existing process or observe similar processes
  - Collect data on the existing operation and document the details in a format that provides itself to examination.
  - Conduct experiments on the process.
  - Develop a mathematical model of the process or utilize an existing mathematical model
  - Perform a computer simulation of the process.
  - Use charting techniques.

Methods Study -2

- Formulate alternatives

  There are always multiple ways to perform a task or accomplish a process, some of which are more efficient and effective than others.

  The purpose of this step is not to identify the best alternative but to formulate all of the alternatives that are feasible.

- Evaluate Alternatives and Select the Best.

  This step consists of a systematic assessment of the alternatives and the selection of the best solution among them, based on the original definition of the problem and objectives.

Standardization

Implement the Best Method:

- introducing the changes proposed in the existing method or operation,
- or installing the new method or process.

This may involve:

- pilot studies or trials of the new or revised method
- training operators to follow the new standard methods
- complete documentation of the new or revised method and
- replacement of the previous documentation

Control

Perform an audit or follow-up on the methods engineering project.

- How successful was the project in terms of the original problem definition and objectives?
- What were the implementation issues?
- What should be done differently in the next methods engineering study?

Techniques of Methods Engineering

- Process Chart Analysis
- Operations Analysis
- Motion study and work design
- Facility layout planning
- Work measurement techniques

Process Chart Analysis

A process chart is a graphic representation of events occurring during series of actions or operations that form a methods.

- Operations process chart
- Flow process chart
- Flow diagram
- Operator process chart or left-hand/right hand chart
- Man and machine chart
- Simo chart
- Gang process chart
- Multi-activity process chart
- Travel chart or from-to-chart

Exercise

These charts are used under different conditions, depending on the investigations or analysis being undertaken.

- The Method Analyst therefore needs to be familiar with all of these charts in order to use them appropriately.

Assignment

Choose two of the charts, write a brief description of them and what they are used for. Also, make a drawing of a typical example and explain the method taking place.

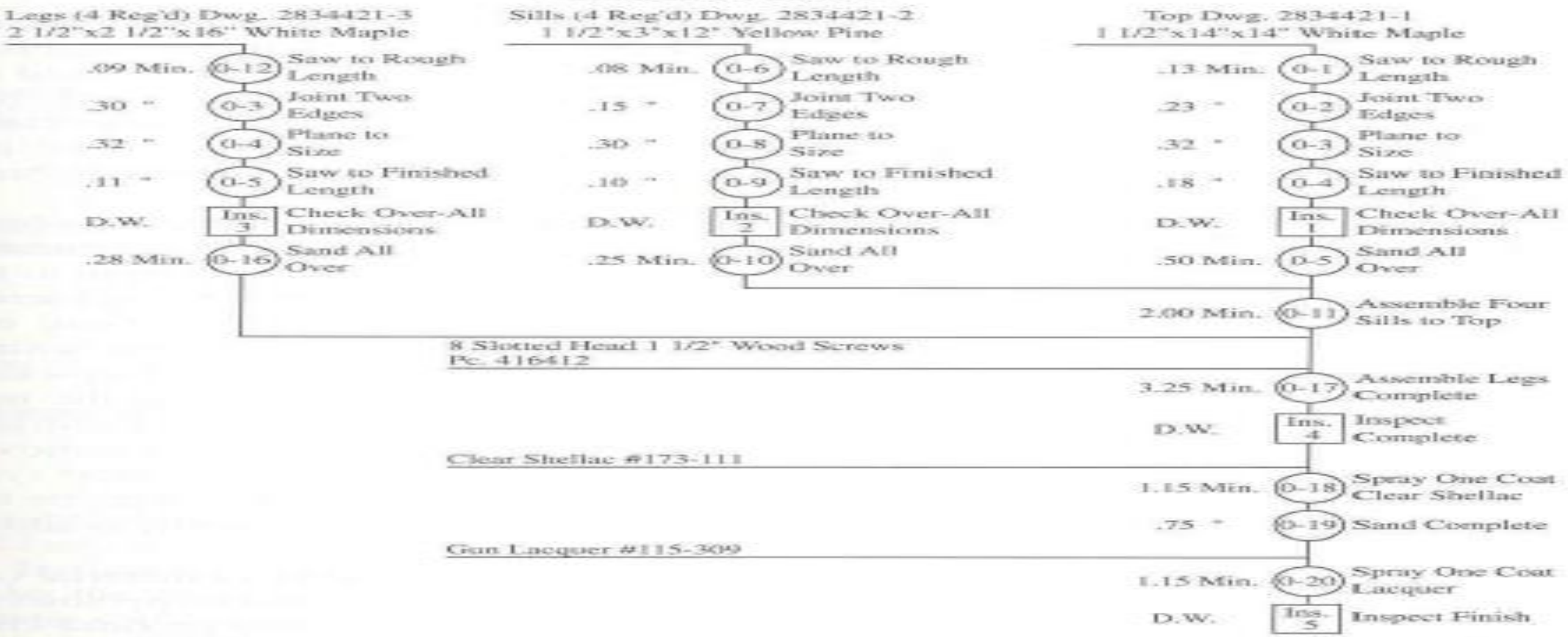
An Example: Operations Process Chart

The operation process chart shows the chronological sequence of all operations, inspections, time allowances, and materials used in a manufacturing or business process, from the arrival of raw material to the packaging of the finished product.

It is particularly useful in connection with plant layout studies.

Operation Process Chart

Manufacturing Type 2834421 Telephone Stands--Present Method
Part 2834421 Dwg. No. SK2834421
Charted By B.W.N. 4-12-



Summary:

Event	Number	Time
Operations	20	17.58 minutes
Inspections	5	Day work

FLOW PROCESS CHART

The flow process chart is a tool for recording a process, with the intent to understand it better and improve it.

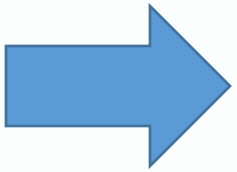
- represents graphically the separate steps or events that occur during the performance of a task or doing a series of actions.
- The chart usually begins with the raw material entering the factory and follows it through every step.



Operation

D

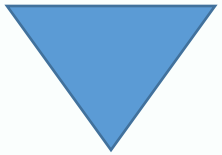
Delay



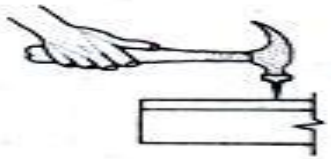
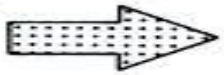
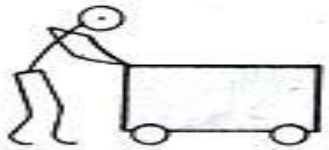
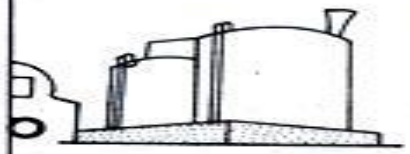
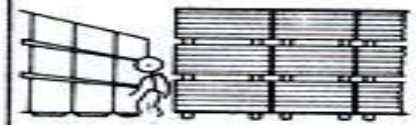
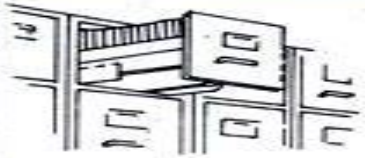
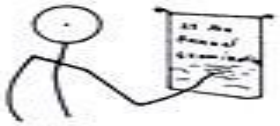
Transportation



Inspection



Storage

Operation  A large circle indicates an operation such as	 Drive nail	 Mix	 Type letter	 Drill hole
Tranportation  An arrow indicates a transportation such as	 Move materal by truck	 Move material by conveyor	 Move material by carrying (messenger)	 Move material by hoist or elevator
Storage  A triangle indicates a storage, such as	 Material in Factory store	 Finished stock stacked on pallets	 Protective filing of documents	 Bulk storage of raw materials
Delay  A large capital D indicates a delay, such as	 Wait for elevator	 Material in truck or on floor at bench waiting to be processed	 Papers waiting to be filed	 Finished product waiting for packaging
Inspection  A square indicates an inspection, such as	 Examine material for quality or quantity	 Read steam gage on boiler	 Examine printed form for information	 Examine weather before going out

Operations Analysis

There are 9 major considerations of operations analysis that are applied in methods study:

1. Purpose of operations
2. Design of part
3. Tolerances and specification
4. Material
5. Process of manufacture
6. Set up tools
7. Plant layout
8. Work conditions, and
9. Motion and time study

Motion Study

Two approaches to motion study:

1. **Macro motion study:** This is the study of the overall product flow through the manufacturing facility
2. **Micro motion study:** This is used to understand individual operations using techniques that enable an understanding of the smallest detail of each operation so improvement can be made.

Macro Motion Study

-Defined as the flow of a product as it moves through the factory.

It makes use of the five process chart symbols we looked at before: delay, operation, transportation, inspection and storage.

Techniques for Macro Motion Study

- Flow diagram
- Operations Chart
- Flow process chart
- Works cell load chart
- Route Sheet

Micro Motion Study

-Looks at the study of individual operations

-Operations or work is broken down into motions like :

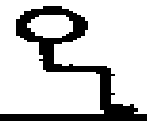
- Reach,
- Move,
- Grasp,
- Position,
- Align etc

-Time it takes to undertake each activity is measured.

Gilbreth concluded that any work can be done using a combination of 17 basic motions, called Therbligs.

The breaking down of operations into individual motions, including delay in the process helps to identify inefficient motions and to utilise or eliminate even split-seconds of wasted time.

Therbligs

	SEARCH		INSPECT
	FIND		PRE-POSITION
	SELECT		RELEASE LOAD
	GRASP		TRANSPORT EMPTY
	TRANSPORT LOADED		REST FOR OVER COMING FATIGUE
	POSITION		UNAVOIDABLE DELAY
	ASSEMBLE		AVOIDABLE DELAY
	USE		PLAN
	DISASSEMBLE		

Some Micro Motion Study Techniques

- Operations analysis chart –used to describe a single activity
- Operator/machine chart – used for two activities
- Gang chart- used when two or more people work together,
- Multi-machine chart
- Left-hand/right-hand chart – for one operator, treats each hand as an activity

NB: Techniques have some factors in common which include:
-each activity is broken down into elements;
-Time is measured linearly in minutes.

All operations analysis techniques can use the same form but the number of activities varies.

Standardization and Operator Training

After standardization, it is imperative that equipment, materials and working conditions be standardized to ensure that the proposed methods is followed at all applicable work centers.

-It is very important to train the operator

-Depending on the nature of the job, training can consist of:

-Verbal instruction demonstrations at or away from the from the workplace,

-Written instruction cards,

-Visual guidance with video tapes or motion pictures,

-Or a combination of these and other training techniques.

Labour Standards and Work Measurements

The number of standard hours in which operators, working on standardised job or working with standardised method is determined by measurement.
Labour standards (time standards) are the amount of time required to perform a job or part of a job.
Labour standards help to determine:

- labour content of items produced (the labour cost)
- Staffing needs of organisations
- Production expected
- Basis of wage-incentive plans
- Cost and time estimates

Labour or time standards will let employers know the amount of time it should take an average employee to perform specific job activities under normal working conditions

Setting Labour or Time Standards

Techniques used for setting time standards include:

- Historical experience (relatively easy and inexpensive to obtain)
- Time studies (Proposed by Taylor and most widely used)
- Predetermined Time Standards (Prohibitively expensive)
- Work sampling

Time Studies

It is often called the classical stop watch study

It involves timing a sample of a worker's performance and using it to establish a standard .

Procedure:

1. Define the task to be studied (method analysis)
2. Break down the task into precise elements (parts of a task that often take no more than a few seconds
3. Decide how many times to measure the task (number of cycles or samples needed)
4. Time and record the elemental times and ratings of performance

Time Studies: Procedure Contd.

5. Compute the average actual cycle time

Average cycle time = sum of each elemental time measured / Number of cycles observed

6. Compute the normal time for each element- it is a measure of a performance rating for a particular worker pace observed (adjusts the observed time to what a normal worker could expect)

Normal Time = Average cycle time * performance rating

7. Sum the **normal times** for each element to develop a **total normal time** for the task.
8. Compute the standard time **(**it takes into account time for personal needs, unavoidable delays and worker fatigue)

Standard time = Total normal time / 1 – allowance factor

Personal allowances are often established in the range of 4% to 7% of total time

Example 1

The time study of a work operation yielded an average actual cycle time of 4.0 minutes. The actual analyst rated the observed worker at 85% . This means the worker performed at 85% of normal time when the study was made. The firm uses a 13% allowance factor. Compute the standard time.

Try these and provide the answers in class

Example 2

Management Science Associates promotes its management development seminars by mailing thousands of individually types letters to various firms. A time study has been done on the task for preparing letter for mailing. On the basis of the observation below, Management Science Associates wants to develop a time standard for the task. The firm's personal delay and fatigue allowance factor is 15%

Job Element	Cycle Observed (in minutes)					Performance Rating
	1	2	3	4	5	
Type letter	8	10	9	21	11	120%
Type envelop address	2	3	2	1	3	105%
Stuff, stamps, seal and sort envelopes	2	1	5	2	1	110%

Determining the Actual Sample Size

Time studies require **sampling** to measure.

Sampling error in the average actual cycle time naturally arises.

- According to statistics, error varies inversely with sample size.

To determine adequate sample size, three items must be considered:

1. How accurate we want to be (+/- 5) of actual be enough?).
2. The desired confidence level (z value; is 95% or 99% required?)
3. How much variation exists within the job elements (i.e. if variation large, a larger sample is required and vice versa).

Formula for Estimating Sample Size

The formula for finding the appropriate sample size given three variables is:

$$n = \left(\frac{zs}{h\bar{x}}\right)^2$$

Where:

h: is the accuracy level desired in percent of the job element, expressed as a decimal (e.g. 5% = 0.05)

z: is the number of standard deviations required for the desired level of confidence (e.g. 90% confidence=1.65 as shown below for the more common values of z.)

s: standard deviation of the initial sample

X is the mean of the initial sample.

Determining the Appropriate Sample Size

Desired Confidence (%)	z-value Standard Deviation required for desired level of confidence
90	1.65
95	1.96
95.4	2
99	2.5

Note, that work sampling technique could also be used to determine actual sample size

Formula for Estimating Sample Size

- Bob Swan Manufacturing has asked you to check a labour standard prepared by a recently terminated analyst. Your first task is to determine the correct sample size. Your accuracy is to be within 5% and your confidence level to be 95%. The standard deviation of the sample is 1.0 and the mean is 3.00.
- Solution:
- h =0.05 $\bar{x}$= 3.00 S =1.0
- z =1.96 (from Normal distribution table above)
- $n = \left(\frac{zs}{h\bar{x}}\right)^2$
- $n = \left(\frac{1.96*1.0}{0.05*3}\right)^2$
- Therefore, you recommend a sample size of 171.

Predetermined Time Standards

- A technique for setting time standards
- It involves **dividing manual work** into small basic elements that have established times (based on very large samples of workers)
- **To estimate the time for a particular task**, the time factors for each basic element are added together
- It is prohibitively expensive

The most common technique is the : **Methods Time Measurement** (MTM)

- Operations are broken down into basic tasks corresponding to **therbligs**.
- Each basic motion is stated in **Measurement Time Units** (MTU)
- To estimate the time it takes to perform a particular **task simply add MTUs**
- MTU's could be converted to **minutes** or **hours** by multiplying by a number
- For minutes conversions multiply by 0.0006
- For hourly conversions multiply by 0.00001

Predetermined Time Standards

Advantages:

- May be established in a laboratory environment (production is not disturbed);
- Standards can be set before a task is done;
- No performance ratings are necessary;
- Methods widely accepted by unions as a fair means for setting standards.

Predetermined Time Standards: An Example

Riveting a transistor board in an assembly process is assigned a **MTU** value of **70 MTU**, based on industry data standards. Before riveting, a worker must reach 165 inches for a small part **(17 MTU)**, grasp the part **(9.1 MTU)**, move the part to the assembly **(27 MTU)**, and position the transistor (**32.3 MTU**). Estimate the time it take to do the whole operation.

Solution:

This very small task, which consists of five elements, takes a total of 155.4 TMU (17.0 + 9.1 + 27.0 + 32.3 + 70.0).

Translating into minutes involves multiplying 155.4 MTU x .0006 minute = .0932 minute = 5.6 seconds.

Work Sampling

A technique **for time studies developed by L. Tippet** in 1930s.

The technique **estimates the percent of time a worker spends working** on various task.

Involves **random observations** to record the activity that the worker is performing.

Work Sampling Applications

- Work sampling is used in:

1. *Ratio delay studies.* These estimate the percentage of time employees spend in unavoidable delays. The results are used to investigate work methods, to estimate activity costs, and to set allowances in labour standards.
2. *Setting labour standards*. For setting standard task times, the observer must be experienced enough to rate the worker's performance.
3. *Measuring worker performance*. Sampling can develop a performance index for workers for periodic evaluations.

Steps for Work Sampling

1. Take a preliminary sample to obtain an estimate of the parameter value (such as percent of time a worker is busy);
2. Compute the sample size required;
3. Prepare a schedule for observing the worker at appropriate times. The concept of random numbers is used to provide for random observations;
4. Observe and record worker activities; rate the worker's performance;
5. Record the number of units produced during the applicable portion of the study;
6. Compute normal time;
7. Compute standard time.

Determining the Number of Observations

To determine the number of observations required, management must make a statement about the desired confidence level and accuracy.

But first the work analyst must select a preliminary value of the parameter under study (step 1 above).

The choice is usually based on a small sample of perhaps 50 observations. The following formula then gives the sample size for a desired confidence and accuracy:

$$n = \frac{z^2 p(1 - p)}{h^2}$$

where,

n = Required sample size

Z = Standard normal deviation for the desired confidence level (Z = 1 for 68% confidence) These values are obtained from the normal table.

p = Estimated value of sample proportion (of time worker is observed busy or idle)

h = Accuracy level desired, in percent

Example: Determining the Number of Observations

The head of a large typing pool estimates that the typists are idle 25% of the time. The supervisor would like to take a work sample that would be accurate within 3% and wants to have 95.45% confidence in the results.

In order to determine how many observations should be taken, the supervisor applies the equation:

$$n = \frac{Z^2 p (1 - p)}{h^2}$$

where,

n = Sample size required

Z = 2 for 95.45% confidence level

p = Estimate of idle proportion = 25% = 0.25

h = Accuracy desired of 3% = .03

It is found that:

$$\frac{(2)^2 (0.25)(0.75)}{(0.03)^2} = 833 \text{ observations}$$

Work Sampling

Work sampling is used to set labour standards in a fashion similar to that used in times studies.

The analyst, however, records whether a worker is busy or idle during the observation

After observations, the worker is rated, and the units produced counted (step 4 and 5)

The normal time can then be estimated.

Work Sampling Formulas

$$\text{Normal time} = \frac{(\text{total study time}) * (\%\text{ of time employee is observed working}) * (\text{performance rating})}{\text{Number of pieces produced}}$$

$$\text{Standard time} = \frac{\text{Total normal time}}{1 - \text{allowance factor}}$$

Work Sampling: An Example

A work-sample study conducted over the 80hrs (or 4800 minutes) of a two week period yielded the following data. The number of parts produced is 225 by an operator who was performance rated at 100%. The operator's idle time was 20%, and the total allowance given by the company for this task is 25%.

1. What is the normal time?
2. What is the standard time?

Working Sampling: Solution

$$normal\ time = \frac{(total \text{ study time}) \text{x } (\% \text{ of time working}) \text{x } (\text{performance rating factor})}{number\ of\ units\ completed}$$

$$normal\ time = \frac{(4{,}800 \text{ min } x0.80x1.00)}{225} = 17.07 \text{ minutes}$$

Standard Time = 22.67 min./part

Time Study Vs. Work Sampling

Work sampling offers several advantages over time study methods.

- it is less expensive, since a single observer can observe several workers simultaneously.
- -observers usually don't require much training, and no timing devices are needed.
- the study can be delayed temporarily at any time with little impact on the results.
- because work sampling uses instantaneous observations over a long period, the worker has little chance of affecting the study's outcome.

The disadvantages of work sampling are:

- it does not break down work elements as completely as time studies,
- it can yield biased or incorrect results if the observer does not follow random routes of travel and observation, and;
- it is less effective than time studies when cycle times are short.